MILITARY SPECIFICATION

ANTENNA ELEMENTS, AS-3435/BRC AND AS-3436/BRC

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, manufacture, and test requirements for Antenna Elements, AS-3435/BRC and AS-3436/BRC, hereinafter referred to as the antenna elements.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standard(s). The following specifications and standard(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-S-760	Strapping, Nonmetallic (And Connectors)

MILITARY

MIL-C-12000	Cable, Cord, And Wire, Electric; Packaging Of
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms And Definitions

2.1.2 Other Government drawings (DWGs) and publications. The following other Government DWGs and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL UNDERWATER SYSTEMS CENTER (NUSC), New London Laboratory (NLL)

NUSC/NLL 02203001	Tip, Cable Termination
NUSC/NLL 02377001	Antenna Element Assemblies, AS-3435/BRC And AS-3436/BRC (for AS-3436/BRC only)
NUSC/NLL 0238006	Assembly Instructions For AS-3434(OV)/BRC
NUSC/NLL 02387001	Adapter, Radio Frequency, MX-10158/BRC

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5985

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PUBLICATIONS

NUSC

NUSC-C-342/4141-279	Cable, Radio Frequency, Buoyant, General Specification

(Copies of specifications, standard(s), DWGs, and publication(s) required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DWGs

SPEARS ASSOCIATES, INCORPORATED (SAI)

SAI D185015	Cover, Test/Shipping
SAI D185049	Housing Assembly
SAI D205020	Termination Tip Grounded
SAI E185017	Connector, Plug, Electrical (Pin Contact) Antenna Assembly Element
SAI E185033	Connector, Plug, Electrical (Socket Contact) Cable Assembly R.F
SAI E211025	Antenna Element Assembly, AS-3435/BRC
SAI E211026	RF Cable Assembly, CG-3820/BRC

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

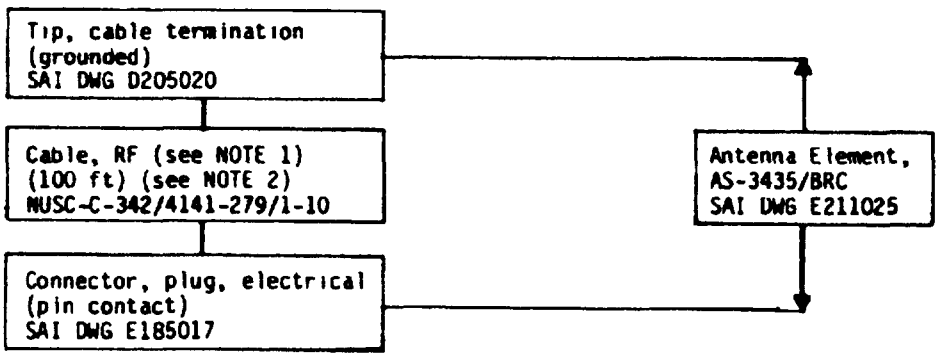
3 REQUIREMENTS

3 1 General. The antenna elements shall be in accordance with MIL-E-16400 to the extent specified herein

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Description. The antenna elements shall each consist of single conductor buoyant cable with length and terminations as specified

3 1 2.1 Antenna Element, AS-3435/BRC Antenna Element, AS-3435/BRC shall consist of an assembly of components as shown in FIGURE 1.



NOTES. 1. Radio frequency
2. Feet

FIGURE 1 Components of Antenna Element, AS-3435/BRC.

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3.1.2.2 Antenna Element, AS-3436/BRC. Antenna Element, AS-3436/BRC shall consist of an assembly of components as shown in FIGURE 2.

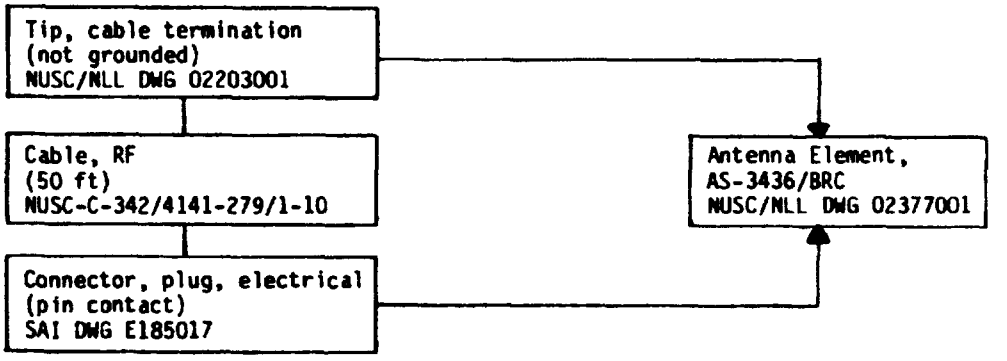


FIGURE 2. Components of Antenna Element, AS-3436/BRC.

3.2 Characteristics. Characteristics of the antenna elements shall be in accordance with 3.2.1 through 3.2.2.8.

3.2.1 Performance. Performance of the antenna elements shall be in accordance with 3.2.1.1 through 3.2.1.3.

3.2.1.1 Antenna Element, AS-3435/BRC. The antenna element shall function as an antenna radiator to receive very low frequency (VLF), low frequency (LF), medium frequency (MF), high frequency (HF), and very high frequency (VHF) communications. The antenna element shall interface with RF Adapter, MX-10158/BRC and RF Amplifier, AM-7074/BRC of Antenna Group, AS-3434(V)/BRC. The antenna element shall be mechanically compatible with deployment and retrieval through Antenna Transfer Assemblies, AN/BRA-18( ) and AN/BRA-24( ).

3.2.1.2 Antenna Element, AS-3436/BRC. The antenna element shall function as an antenna radiator to transmit MF and HF at levels up to one kilowatt (kW) and receive at MF and HF frequencies. The antenna element shall interface with the RF Adapter, MX-10158/BRC of Antenna Group, AS-3434(V)/BRC. The antenna element shall be mechanically compatible with deployment and retrieval through Antenna Transfer Assemblies, AN/BRA-18( ) and AN/BRA-24( ).

3.2.1.3 Direct current (DC) resistance. When tested in accordance with 4.5.1, the antenna elements shall conform to the continuity and insulation resistance limits specified in TABLE I.

TABLE I. Antenna element DC resistance requirements.

Measurement points (see FIGURE 3 and FIGURE 4)	DC resistance	
	AS-3435/BRC	AS-3436/BRC
A and C (continuity)	Less than 0.42 ohms	Less than 0.21 ohms
B and C (insulation)	Greater than 1000 megohms	Less than 1000 megohms

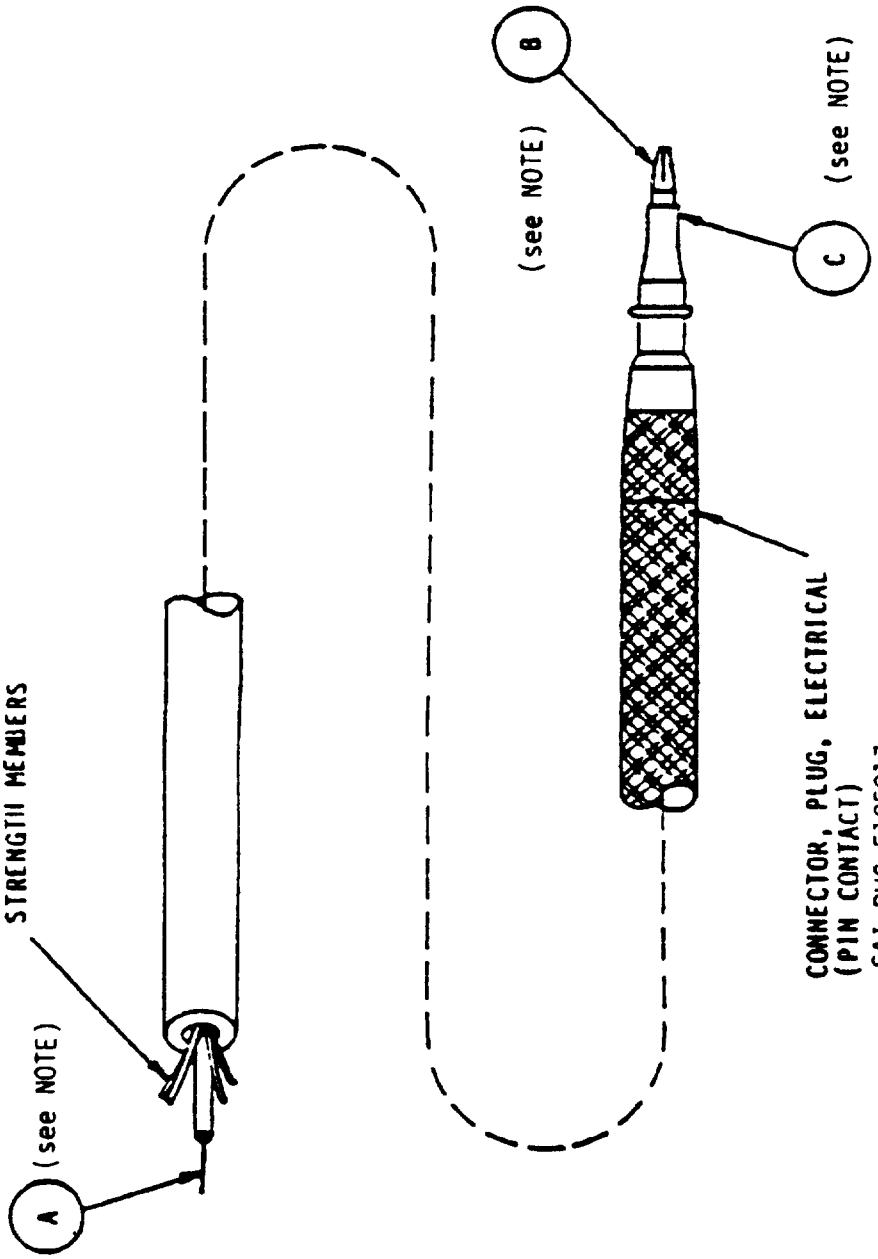
3.2.2 Physical characteristics. The antenna elements shall be fabricated in accordance with this specification and the guidance shown on SAI DWG E211025.

3.2.2.1 Flexibility. When tested in accordance with 4.5.7, the antenna elements shall not exhibit physical damage as revealed by surface examination and pressure test, and electrical degradation as revealed by the DC resistance test.

3.2.2.2 Pressure resistance. When tested under hydrostatic pressure in accordance with 4.5.2, the antenna elements shall not show evidence of damage or flooding.

3.2.2.3 Durability. The antenna elements shall withstand the AN/BRA-24( ) durability test of 4.5.10. The antenna elements shall not exhibit degradation as revealed by the surface examination, the DC resistance test, and the marking examination.

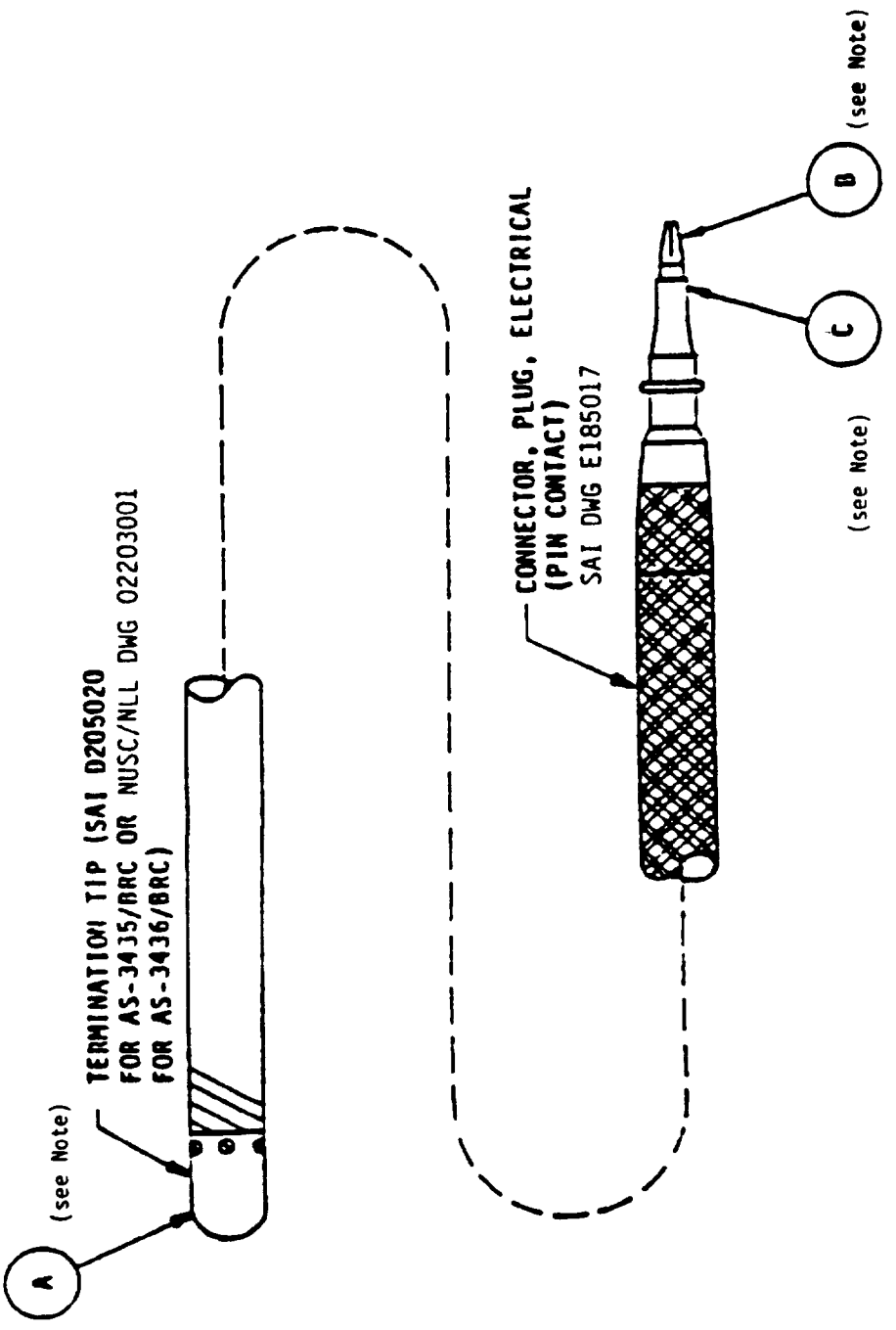
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NOTE:
1. Measurement point for antenna element DC resistance.

FIGURE 3. DC resistance test points for partially assembled antenna element.

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Note 1. Measurement point for antenna element DC resistance

FIGURE 4. DC resistance test points for antenna element.

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3.2.2 4 Tensile strength. When tested in accordance with 4.5.9 the molded connector plug and the molded termination tip (with suitable test fixtures attached), shall withstand static tensile loads along the axes of the connector plug and the termination tip with no evidence of cracking or other damage. The maximum elongation of the molded section, including the separation of connector or termination tip (metal sections) from the buoyant cable, shall not exceed 2 percent when subjected to tensile loads as specified in 4.5.9.

3.2.2.5 Temperature nonoperating. The antenna elements shall withstand shipment and storage temperatures over the range -40° Celsius (C) to +70°C.

3.2.2.6 Temperature operating. When tested in accordance with 4.5.8.2, the antenna elements shall operate at a temperature of -10°C.

3.2.2.7 Diameter. The antenna elements shall be fabricated from buoyant cable which conforms to NUSC-C-342/4141-279. The outer diameter of any point along the antenna element, including the molded connector and termination tip, shall be 0.650 inches (in.) ±0.020 in.

3.2.2.8 Length. Unless otherwise specified (see 6.2), the overall length of the AS-3435/BRC and AS-3436/BRC antenna elements shall be 100 feet (ft) ±1 ft and 50 ft ±1 ft, respectively. The antenna elements shall be continuous without joints or splices and shall be tested in accordance with 4.5.6.

3.3 Health and safety. Health and safety considerations require avoidance of toxicological substances such as encapsulants, adhesives, or preservatives which could contaminate the closed environment of a submarine. The materials specified in SAI DWG E211025 shall be acceptable Substitutes shall be approved by the Government.

3.4 Marking. The antenna elements shall conform to the marking and identification requirements of DWGs, part lists, specifications, and other documents specified on SAI DWG E211025.

3.5 Workmanship. The antenna elements shall be manufactured with care using the exact materials, dimensions, and surface finishes specified. Correct arrangement, hookup, and fastening of all components during assembly shall be assured. The electrical molding procedure for the connector plug and termination tip is particularly critical and shall be performed under controlled time and temperature conditions. The completed antenna elements shall be clean and free of sand, grit, burrs, cuts, and gouges.

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

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4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   1. Production inspection (see 4.4.1)
   2. Production control inspection (see 4.4.2)
c. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

TABLE II. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
DC resistance	3.2.1.3	4.5.1	X	X	
Pressure resistance	3.2.2.2	4.5.2.1, 4.5.2.2	X	X	
Physical characteristics	3.2.2	4.5.3	X	X	
Marking	3.4	4.5.4	X	X	
Diameter	3.2.2.7	4.5.5	X	X	
Length	3.2.2.8	4.5.6	X	X	
Flexibility	3.2.2.1	4.5.7	X	X	
Temperature nonoperating	3.2.2.5	4.5.8.1	X		X
Temperature operating	3.2.2.6	4.5.8.2	X		X
Tensile strength	3.2.2.4	4.5.9	X		X
Durability	3.2.2.3	4.5.10	X		
Workmanship	3.5	4.6	X	X	

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with 4.4.1 through 4.4.2.2.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each antenna element offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE II, Group A.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operation, and tests to determine deviations in the processes and equipment required to fabricate the antenna elements. These tests shall be performed on the complete antenna element as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE II, Group B.

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4.4.2.1 Sampling plan for production control inspection. Antenna elements submitted for production control sampling tests shall be selected by the Government Quality Assurance Representative (QAR) and shall be representative of current production and shall have passed production inspection specified in 4.4.1. One antenna element shall be selected from each successive 50 production antenna elements (AS-3435/BRC or AS-3436/BRC), or fraction thereof, produced, using a single run of buoyant RF cable conforming to NUSC-C-342/4141-279.

4.4.2.2 Production control sample unit processing. If a sample unit fails the inspection specified in 4.4.2.1, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. No sample shall be delivered as a production item under the contract.

4.5 Test methods. To the extent specified herein, the examinations and tests specified in TABLE II shall be performed in accordance with the test methods specified in MIL-E-16400. Testing shall be performed at ambient pressure and room temperature unless otherwise specified.

4.5.1 DC resistance test The antenna elements, partially assembled as shown in FIGURE 3, or completely assembled as shown in FIGURE 4, shall be tested for continuity using a General Radio Type 1650-A impedance bridge, or equal. Insulation resistance readings shall be performed using a megohmeter (AN/PSM-25, or equal) which applies not more than 500 volts (V) to the circuit under test. The continuity and insulation resistance limits shall be in accordance with TABLE I. When conducting the resistance measurement between points A and B, after the hydrostatic pressure test (see FIGURE 3), the strength members shall be isolated from the center conductor of the cable; if not, a parallel path will be provided between points A and B resulting in an erroneous reading.

4.5.2 Pressure resistance test. The pressure resistance test shall be in accordance with 4.5.2.1 and 4.5.2.2.

4.5.2.1 Antenna Element, AS-3435/BRC. The antenna element without the termination tip installed shall be tested for watertight integrity, using the test setup shown in FIGURE 5, in accordance with a through i:

a. Place the cover manufactured in accordance with SAI DWG D185015 over the connector plug (pin contact) conforming to the SAI DWG E185017 portion of the partially assembled antenna element.

b Subject the connector plug and cover to hydrostatic pressure of 600 pounds per square inch (psi) as specified in TABLE III with the unterminated end of the antenna element in air at ambient pressure The time to reach the required pressure should be the minimum permitted by the equipment

c. At the end of each pressure test interval, measure the insulation resistance using a megohmeter (AN/PSM-25, or equal) that applies no more than 500 V between test points. Measure between the center conductor (unterminated end of partial assembly) and the ground point on the pressure tank (see FIGURE 5). The insulation resistance readings shall be in accordance with TABLE I.

d. Remove the partial antenna element assembly from the pressure vessel and dry off water on surface of the cable, connector, and cover.

e Perform the surface examination in accordance with 4.5.3.

f. Complete the assembly of the antenna element by installing the molded termination tip, SAI DWG D205020. (The molded termination tip is not hydrostatically pressure tested after assembly.)

g. Temporarily remove the cover placed over the connector plug during a.

h. Perform the DC resistance test of 4.5.1 on the antenna element (see FIGURE 4).

i. Reinstall the cover removed in g on connector plug for protection during shipment (see 5.1).

4.5.2.2 Antenna Element, AS-3436/BRC. The Antenna Element, AS-3436/BRC, without the termination tip installed, shall be tested for watertight integrity, using the test setup shown in FIGURE 5, in accordance with a through l:

a Place the cover manufactured in accordance with SAI DWG D185015 over connector plug (pin contact) conforming to the SAI DWG E185017 portion of the partially assembled antenna element.

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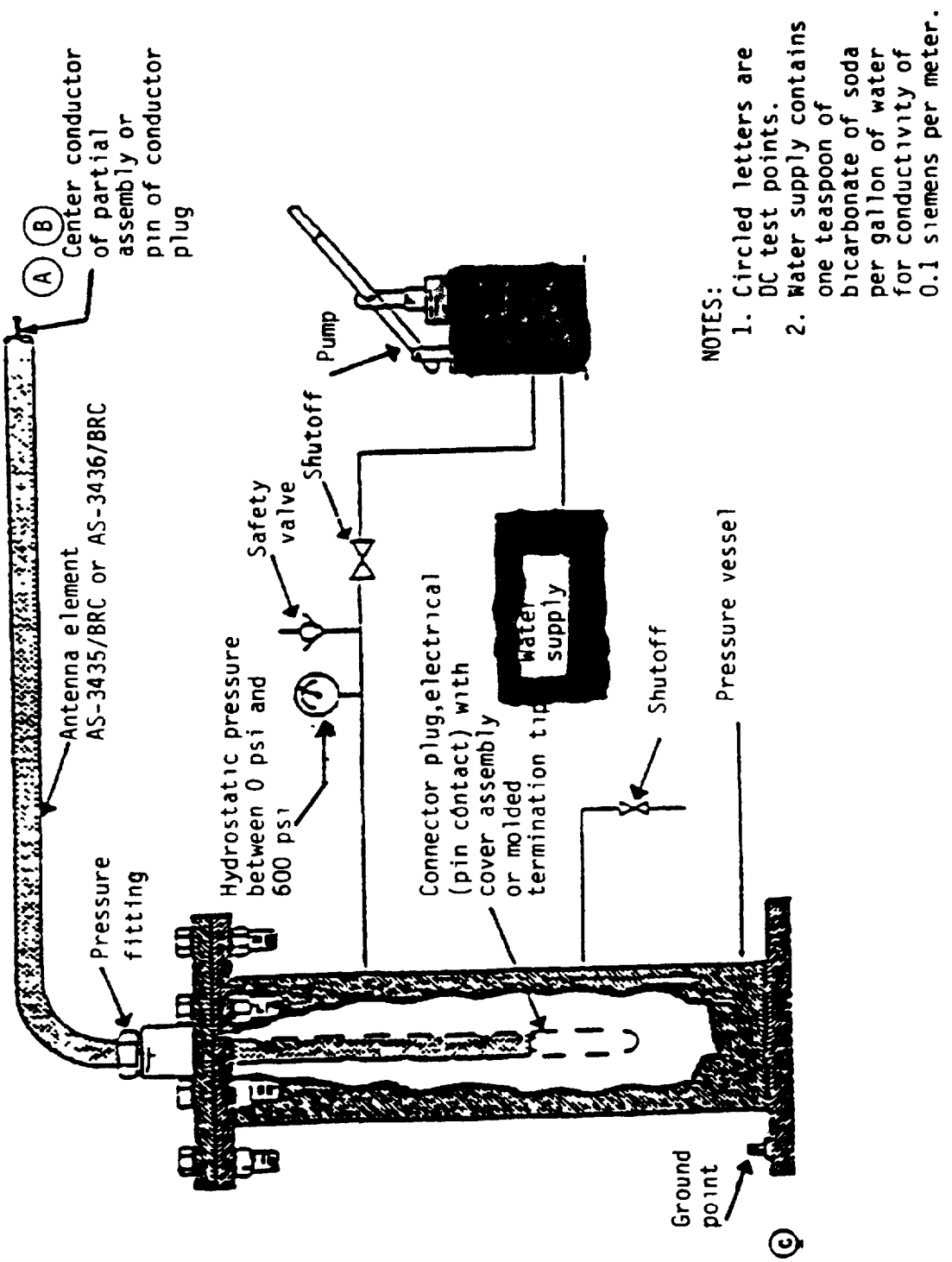


FIGURE 5. Pressure resistance test setup.

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b. Subject the connector plug and cover to a hydrostatic pressure of 600 psi as specified in TABLE III with the unterminated end of the antenna element in air at ambient pressure. The time to reach the required pressure should be the minimum permitted by the equipment.

c. At the end of each pressure test interval, measure the insulation resistance using a megohmeter (AN/PSM-25, or equal) that applies no more than 500 V between test points. Measure between the center conductor (unterminated end of the partial assembly) and the ground point on the pressure tank (see FIGURE 5). The insulation resistance shall be in accordance with TABLE I.

d. Remove the partial antenna element assembly from the pressure vessel and dry off water on the surface of the cable, connector, and cover.

e. Perform the surface examination in accordance with 4.5.3.

f. Complete the assembly of the antenna element by installing the molded termination tip conforming to NUSC/NLL DWG 02203001.

g. Place the termination tip end of the antenna element in the pressure vessel. Subject the termination tip to a hydrostatic pressure of 600 psi, as specified in TABLE III, with the connector plug end of the antenna element in air at ambient pressure. The time to reach the required pressure should be the minimum permitted by the equipment.

h. Temporarily remove the cover placed over the connector plug during a.

i. Initially, and at twenty minutes thereafter (or when the pressure vessel has reached the required pressure), measure the insulation resistance using a megohmeter (AN/PSM-25, or equal) that applies no more than 500 V between test points. Measure between the center conductor of connector plug (pin contact) and the ground point on the pressure vessel (see FIGURE 5). The insulation resistance shall be in accordance with TABLE I.

j. Remove the termination tip from the pressure vessel and dry off the water on the surface of the cable and termination tip.

k. Perform the surface examination in accordance with 4.5.3.

l. Reinstall the cover removed in h on the connector plug for protection during shipment (see 5.1).

TABLE III. Pressure resistance test requirements.

First article		Production	
Pressure (psig) 1/	Duration (minutes)	Pressure (psig)	Duration (minutes)
0	5	0	5
600	25	600	25
0	5	0	5
600	25	600	25
0	5	0	5
600	25	600	25
0	5	0	5
600	20 hours	600	25
0	5	0	5
600	25	600	25
0	5	0	5
600	25	600	25
0	5	0	5
		600	25
		0	5
		600	25
		0	5

1/ Pound-force per square inch gage

4.5.3 Surface examination. Surface examination shall be performed in accordance with MIL-E-16400. Conformance to the requirements of 3.2.2 and 3.3 shall be verified during this examination.

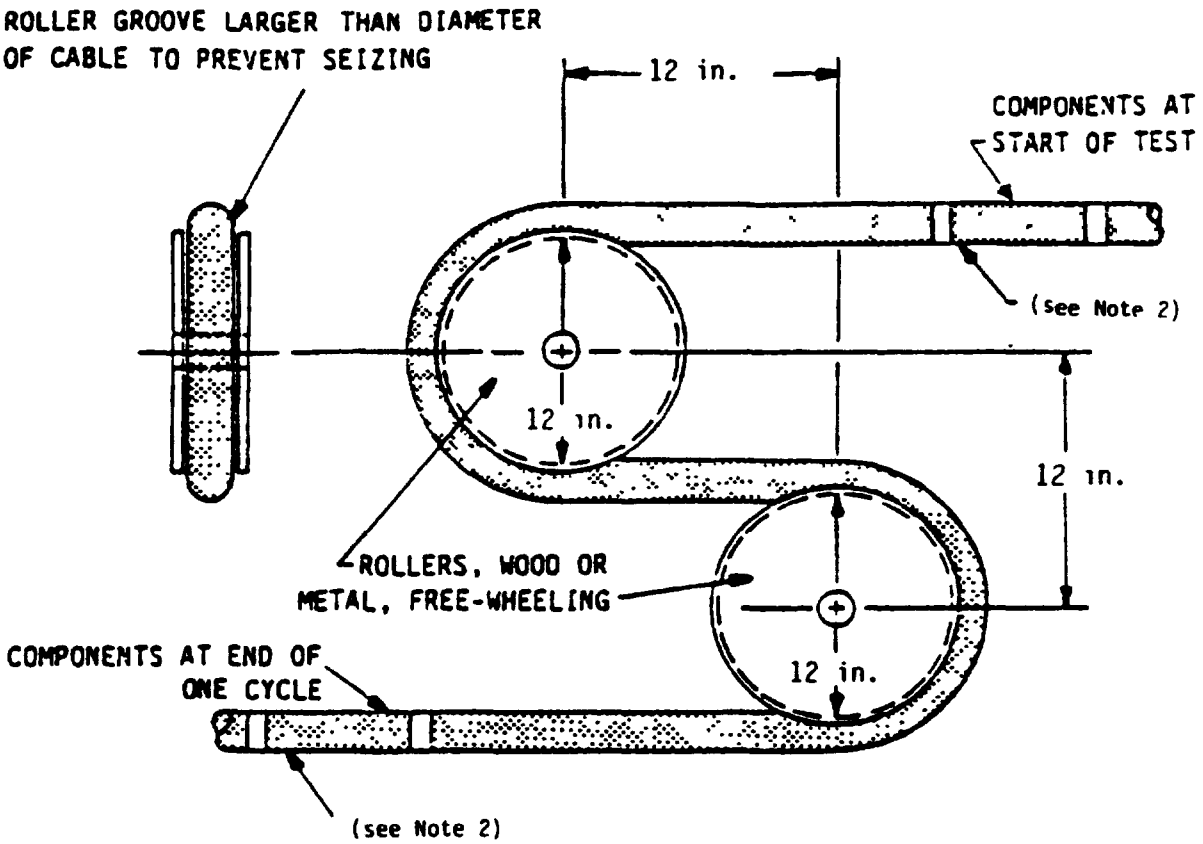
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4.5.4 Marking examination. The marking examination shall consist of verifying that the position marks and identification letters are legible at a distance of 3 ft and that these marks and letters are an integral part of the cable in accordance with 3.4.

4.5.5 Diameter test. The complete antenna element shall be passed through a sizing block having a hole diameter of 0.670 in. to 0.671 in. This test can be run concurrently with the length test of 4.5.6.

4.5.6 Length test. The length of the antenna elements shall be measured by a cable counter or other suitable device.

4.5.7 Flexibility test. The antenna elements, when assembled with mating component, RF Adapter, MX-10158/BRC, and test cable including connector plug (socket contact), shall be flexed through an S-bend path in a test device made in accordance with FIGURE 6, three times in each direction, at 60 ft per minute. The RF adapter and test cable assembly, as shown in FIGURE 7, should be attached to the antenna element to facilitate testing. For the AS-3436/BRC only, the flexibility test shall also be performed on the molded connector plug (pin contact) prior to assembly of the termination tip. The continuity from the center conductor of the test cable to the termination tip of the antenna element (AS-3435/BRC and partially assembled AS-3436/BRC only) shall be monitored continuously during the test. At the conclusion of the test, perform the DC resistance test of 4.5.1.



NOTES: 1. Use sufficient tension to bend antenna elements into close contact with rollers. Ensure that termination tip and connector plug, electrical (pin contact) are held against rollers, making contact through at least 180 degrees of each roller

2. The adapter and test cable assembly shown on FIGURE 7 should be attached to the antenna element to facilitate testing

FIGURE 6. Flexibility test setup.

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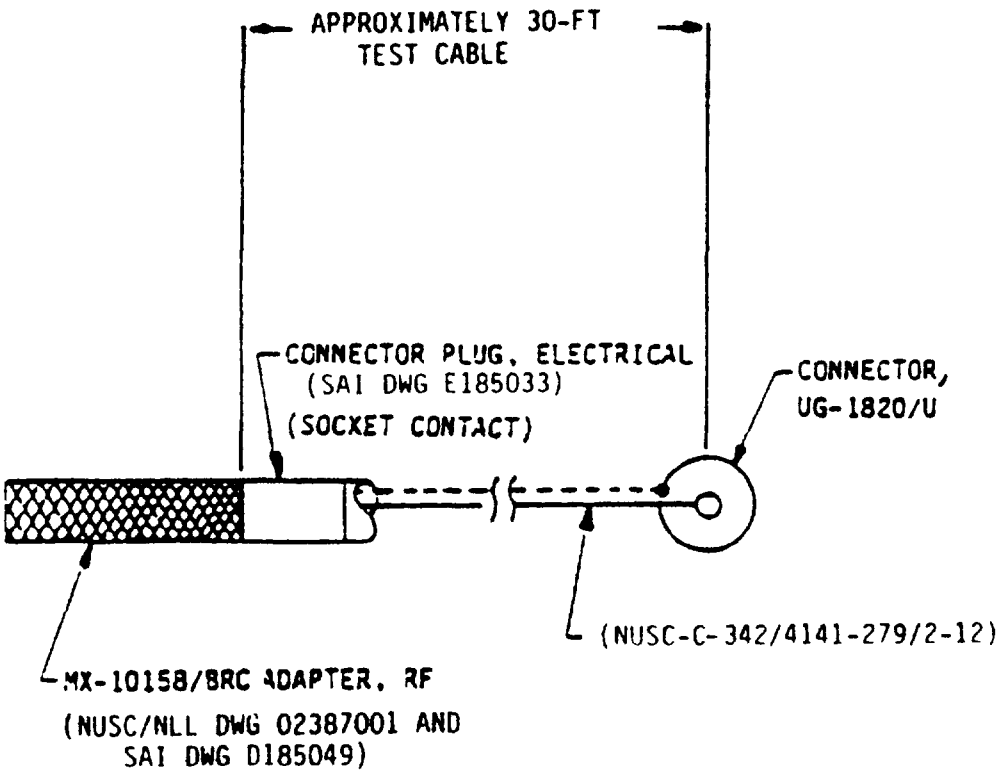


FIGURE 7 RF Adapter, MX-10158/BRC, and test cable assembly.

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4.5.8 Temperature test. Temperature test shall be as specified in 4.5.8.1 and 4.5.8.2.

4.5.8.1 Nonoperating temperature. The antenna element, formed into a coil, with maximum diameter of 22 in. and minimum diameter of 15 in., shall be placed in an environmental chamber and conditioned for 8 hours at -40°C. Increase the temperature at a rate of 10°C per hour until the chamber reaches a temperature of +70°C. Hold the temperature of the environmental chamber at 70°C for 8 hours then remove the antenna element and inspect for damage. The antenna element shall show no evidence of cracking or other damage. Acceptable antenna element performance shall be verified by conducting the flexibility test of 4.5.7.

4.5.8.2 Operating temperature. The antenna element, formed into a coil not over 36 in. in diameter, shall be conditioned in an environmental chamber for 12 hours at -10°C and, while still in the environmental chamber at -10°C, shall be flexed through an S-bend path (as shown in FIGURE 6) three times in each direction at 60 ft per minute. The continuity of the antenna element (AS-3435/BRC only) shall be monitored during the test. Acceptable antenna element performance shall be verified by inspecting the assembly for evidence of damage and by conducting the DC resistance tests of 4.5.1.

4.5.9 Tensile test. The tensile strength of the molded connector plug and the molded termination tip of the antenna element shall be tested by subjecting those portions of the antenna element to static tensile loads along their axes in accordance with TABLE IV. The test setup shall be similar to the setup shown in FIGURE 8. The distance between point A (interface between metal part of connector plug or termination tip and molded section of cable) and measurement point B (six in. from point A) shall be measured under load at the beginning (initial 10-pound (lb) tensile load), and at the 400-lb (connector plug) and 300-lb (termination tip) tensile loads during the fourth cycle. The elongation shall comply with the requirements of 3.2.2.4. The test sample shall be examined for any evidence of cracking or other damage. At the conclusion of the test, perform the flexibility test of 4.5.7.

TABLE IV. Tensile load test requirements.

Connector plug (pin contact)		Termination tip (grounded, ungrounded)	
Tensile load (lb)	Duration (minutes)	Tensile load (lb)	Duration (minutes)
10	Time required to perform reference measurement	10	Time required to perform reference measurement
400	55	300	55
10	5	10	5
400	55	300	55
10	5	10	5
400	55	300	55
10	5	10	5
400	55	300	55
10	5	10	5

4.5.10 AN/BRA-24( ) durability test. The antenna element shall be drawn 25 times in each direction through the Antenna Transfer Assembly, AN/BRA-24( ), using the test setup shown in FIGURE 9. The transfer assembly shall contain water at a hydrostatic pressure of 50 psig to 120 psig. After completion of passage through the transfer assembly, a surface examination, the DC resistance test shall be performed in accordance with 4.5.1 and the marking examination shall be performed in accordance with 4.5.4.

4.6 Workmanship. The antenna elements shall be examined to determine conformance to the surface examination requirements of 3.5 and the Workmanship paragraph of MIL-E-16400.

4.7 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5

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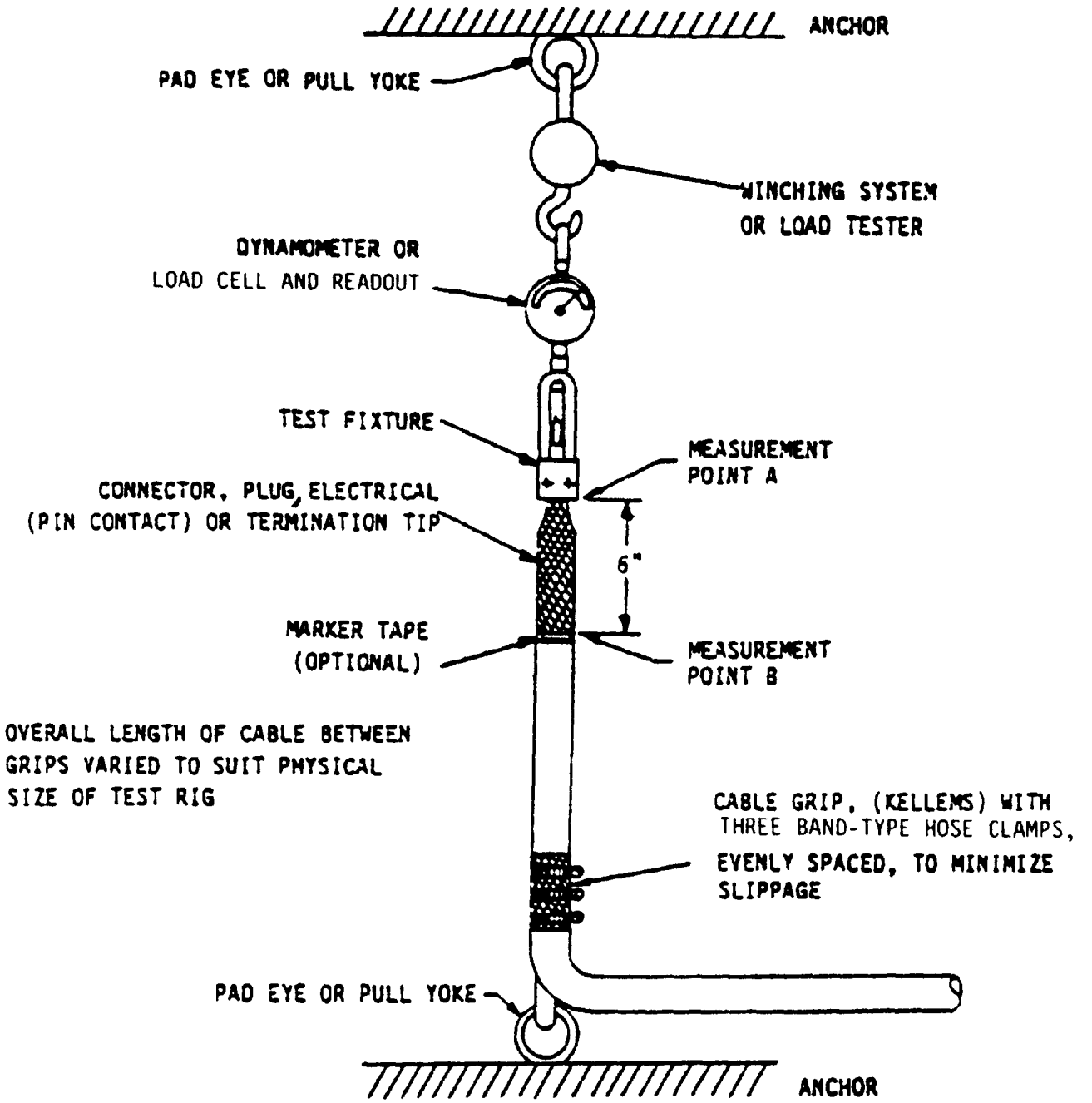


FIGURE 8. Tensile test setup.

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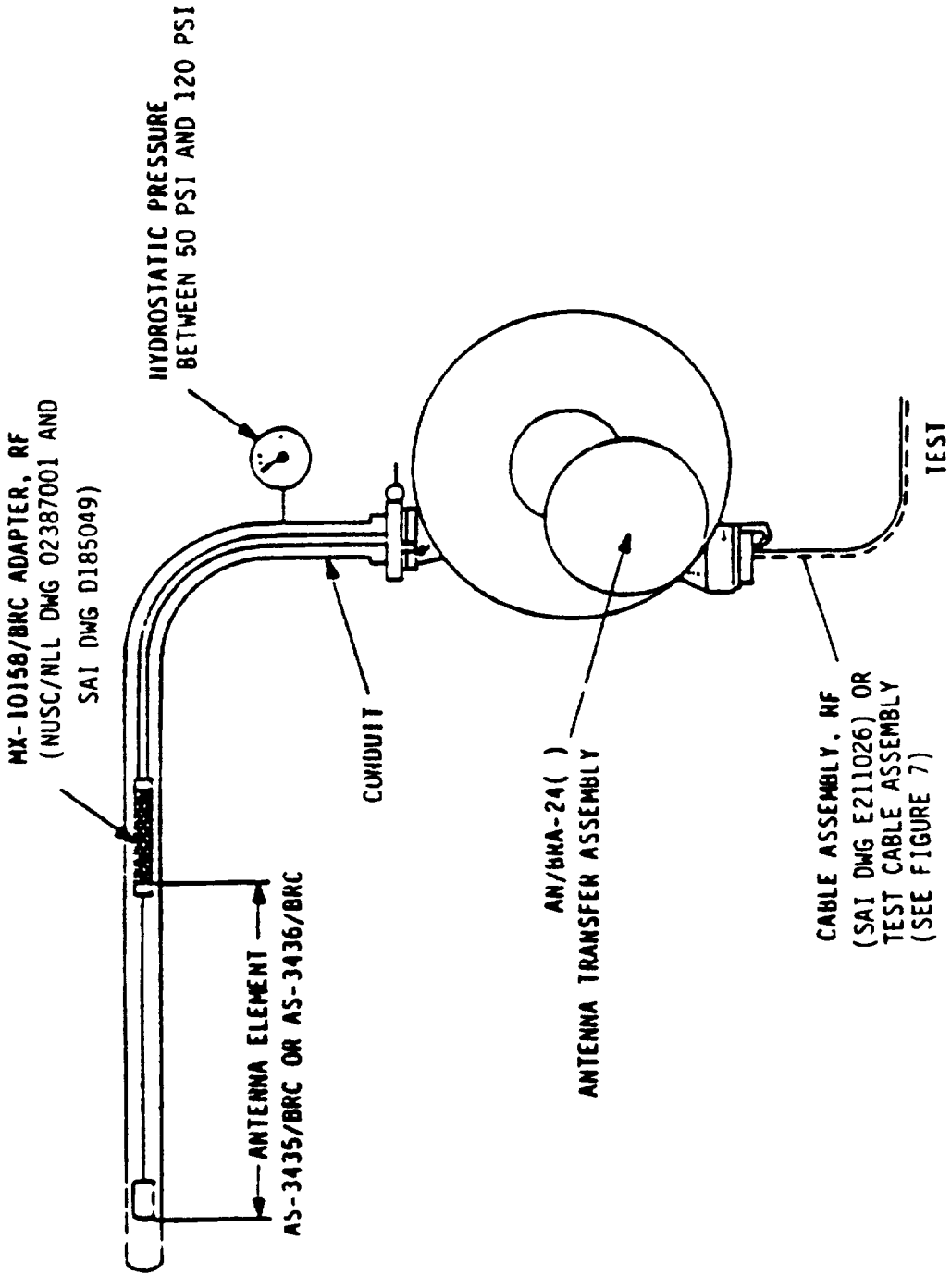


FIGURE 9. AN/BRA-24( ) durability test setup.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. No preservative shall be applied to the antenna elements. Each antenna element shall be shipped in a coil with an inside diameter greater than 15 in. Packaging shall be in accordance with level A of MIL-C-12000 except wood lagging shall be nominal 1 in. thick and shall be strapped with two 0.500 in. by 0.020 in. nonmetallic straps conforming to PPP-S-760, Type II, to afford protection against physical damage during shipment, handling, and uncovered storage. The box containing the coiled antenna element shall fit through a 25-in diameter submarine hatch. A copy of NUSC/NLL DWG 0238006 shall be included in each shipping container. Required connector and termination tip screws, plus a minimum of 50 percent spare screws, shall be packed with each antenna element.

6. NOTES

6.1 Intended use. The antenna elements covered by this specification are used by submarines as part of Buoyant Cable Antenna Assembly, AS-3434(V)/BRC.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4 3
c. Antenna element length, if other than specified in 3.2.2.8

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs

Paragraph	Data requirement	Applicable DID
4.3	First Article Inspection Procedure	DI-T-4901
4.3	First Article Inspection Report	DI-T-4902

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Supersession data. This specification includes the requirements of ELEX-A-524, dated 18 August 1982.

Preparing activity:
NAVY - EC

(Project No. 5985-N525-1(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28895	Antenna Elements, AS-3435/BRC and AS-3436/BRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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